

The friction on a quantized vortex in a truncated Gross–Pitaevskii field follows the temperature, not the normal density

A pre-registered test of $\alpha \propto \rho_n$ against the cutoff, and the correction of a published reading

Xavier Callens

SocrateAI-Scientific-QuantumFluids

<https://github.com/xaviercallens/SocrateAI-Scientific-QuantumFluids>

October 2026

Abstract

The mutual friction α that drags a quantized vortex through the thermal component of a two-dimensional Bose gas is commonly taken to scale with the normal fraction ρ_n/ρ . In an earlier paper [1] we measured $\alpha = 0.0062, 0.014, 0.02$ at three temperatures of the projected Gross–Pitaevskii equation at one cutoff and found $\alpha \simeq 0.24 \rho_n/\rho$; we read the coefficient as a mode-averaged transport cross-section of the vortex, 1.4 healing lengths, and registered the question of whether it depends on the cutoff, with the Born phonon-scattering law as the rival ($\propto k_c$). We now measure α at two further cutoffs, $k_c\xi = 2\pi/3$ and 2π , with the same energy-conserving instrument, with its $T = 0$ known answer reproduced at each cutoff to 0.3 %. The registered prediction of a cutoff-independent coefficient fails, and so does its rival: the coefficient $c = \alpha/(\rho_n/\rho)$ is $0.31 \pm 0.03, 0.232 \pm 0.014$ and 0.102 ± 0.011 at $k_c\xi = 2.1, 3.1, 6.3$, falling by a factor 3.1 while $\rho_n/(\rho T)$ rises by a factor of three to four. What does not move is α itself: $\alpha/T = 0.054 \pm 0.003$ in six arms ($\chi^2 = 1.3$ for 5 degrees of freedom) over three cutoffs and $T = 0.10$ – 0.22 . The proportionality to ρ_n reported before was a degeneracy of a single cutoff, where $\rho_n \propto T$; the mode-averaged cross-section is not a property of the vortex (1.9, 1.4, 0.59 healing lengths) and the “geometric core size” reading is withdrawn. The temperature law was found after the data were read; it was registered as a hypothesis, with its prediction for the next arm fixed in advance: $\alpha = 0.0094 \pm 0.0010$ at $k_c = 2\pi$, $T = 0.173$, against 0.0077 if the friction followed ρ_n . The arm gives 0.0096 ± 0.0018 , inside the window registered for the temperature law — a confirmation by the rule, although the error is almost twice the one assumed when the window was written, so the arm alone separates the two readings by one standard deviation; the discrimination rests on the comparison across cutoffs.

1 Introduction

The mutual friction on a vortex line in a superfluid is parametrised, in the Hall–Vinen form, by α (along the relative velocity) and α' (across it). The standard microscopic picture, from phonon and roton scattering, makes the force proportional to the density of excitations and hence $\alpha \propto \rho_n$ [2, 3]; measurements of the thermal friction on vortices in atomic condensates give values of order 10^{-2} [4, 5], and the only first-principles values for a truncated Gross–Pitaevskii field come from vortex-pair dynamics in the Galerkin-truncated equation [6].

In the preceding paper [1] we measured $\alpha(T)$ in the *projected* Gross–Pitaevskii equation (PGPE), microcanonical, with no damping term and no noise, with estimators that are identities of the dissipative point-vortex model. The result was $\alpha = 0.0062(4), 0.014(2), 0.02(1)$ at $T/T_{\text{BKT}} = 0.14, 0.27, 0.43$, and the ratio $\alpha/(\rho_n/\rho) = 0.23, 0.26, 0.22$. That ratio is the observation this paper tests. We had added an interpretation: for any isotropic bath the drag and

the Landau normal density are sums over the same modes with the same weight (Appendix B), so $\alpha/(\rho_n/\rho)$ is the w -weighted mean of $c_g\sigma_{\text{tr}}$ over the modes, which we evaluated at 1.4 healing lengths, “the geometric size of the core”, and we registered the cutoff dependence as the open question: if the coefficient is a property of the vortex it should not depend on the cutoff that defines the bath; the Born law $\sigma_{\parallel} = \kappa^2 k/8c^2$ [3], extrapolated to the cutoff, predicts $c \propto k_c$.

Registered before any run (PGPE_FRICTION_LAW_PREREG.md, and the kinetic note of the same day): **FL1**, $\alpha \propto \rho_n$ at each cutoff; **FL2**, the coefficient $c = \alpha/(\rho_n/\rho)$ differs between three cutoffs by less than 20 % (the rival: a shift above 40 % growing with k_c); **FL3**, a spread of 10–30 %, ordered $c(2\pi/3) \leq c(\pi) \leq c(2\pi)$, and the Born scaling 0.67 : 1 : 2 as the sharp alternative.

2 Instrument and data

The instrument is that of the preceding paper: PGPE on a periodic box $L = 64$ (units $\hbar = m = g = 1$, mean density 1), projector $|k| \leq k_c$, symplectic split-step evolution with energy drift below 10^{-4} ; antiparallel vortex pairs imprinted with the corrected phase-and-Bernoulli-amplitude profile in thermal base states; sub-grid vortex tracking; α from the energy estimator $\alpha = -\Delta H/(2 \int \sum |v_s|^2 dt)$ and from a two-coefficient regression of the velocities; block jackknife errors; runs end by annihilation or track loss and a validity cut $d \geq 4$ is applied. The cutoffs are $k_c = \pi$ (the published arm, grid $N = 128$), $k_c = 2\pi/3$ (the same grid, projector at a third of the grid wavenumber) and $k_c = 2\pi$ (grid $N = 256$, $dx = \xi/4$).

Base states. Each base is the equilibrated vortex-free $e = 0.60$ state re-projected (and Fourier-resampled) under the new cutoff, heated to a target energy by the phonon construction, evolved 1000 time units, and measured over 500 (temperature from the equipartition fit, $n_s/n = 1 - J_T/J_L$, raw vortex count); a base is admitted only if the vortex count is below 0.5 (Table 1). The fine-grid base at $T = 0.217$ (1.32 vortices) failed admission; the second fine temperature is therefore $T = 0.173$, not the registered 0.22, and it is the arm that was still running when the temperature law was formulated.

cutoff	grid	e	T	n_s/n	ρ_n/ρ	raw n_v (admitted)
$2\pi/3$	$N = 128$	0.545	0.100	0.9846	0.0154	0.00
$2\pi/3$	$N = 128$	0.60	0.216	0.9603	0.0397	0.00
π (published)	$N = 128$	0.60	0.115	0.973	0.027	0.00
π (published)	$N = 128$	0.70	0.220	0.947	0.053	0.00
2π	$N = 256$	0.95	0.127	0.9291	0.0709	0.04
2π	$N = 256$	1.10	0.173	0.9198	0.0802	0.08
2π	$N = 256$	1.25	0.217	0.9029	0.0971	1.32 (<i>not admitted</i>)

Table 1: Base states of the campaign. At comparable T the normal fraction grows steeply with the cutoff ($\rho_n/\rho = 0.015, 0.027, 0.071$ at $T = 0.100, 0.115, 0.127$): the Landau sum runs over more modes.

Known answers. At $T = 0$ (uniform condensate, antiparallel $d_0 = 10$, 400 time units) the pair moves at 0.1065, 0.1068, 0.1065 ($k_c = \pi, 2\pi/3, 2\pi$) and the separation of the four-vortex configuration oscillates between 9.7 and 10.6 with the same phase at all three cutoffs (Fig. 1); $|\hat{\alpha}| \leq 5 \times 10^{-4}$ and $1 - \hat{\alpha}' = 0.9986, 0.9995, 0.9991$. The residual diffusion floor is 2.9×10^{-5} at $2\pi/3$ against a registered clause of 2×10^{-5} : marginal, reported, and a factor 10^2 below the thermal values used below.

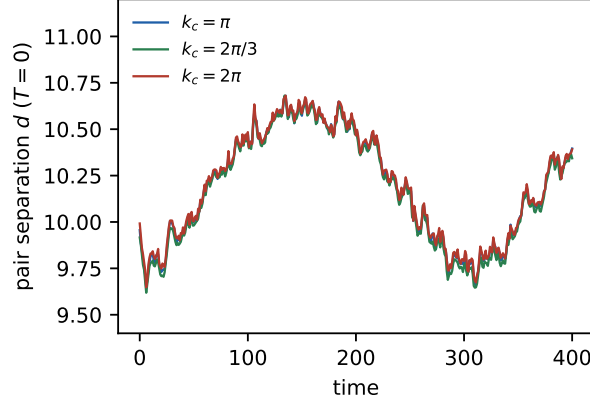


Figure 1: Known answer: the $T = 0$ pair separation at the three cutoffs. Speed 0.1065/0.1068/0.1065; the curves overlies to 0.5 %.

3 Registered predictions and what happened

arm (k_c, T)	runs	ρ_n/ρ	α (energy)	$c = \alpha/(\rho_n/\rho)$	α/T	α'
$2\pi/3, 0.100$	6	0.0154	0.0049 ± 0.0008	0.32 ± 0.05	0.049 ± 0.008	-0.017 ± 0.001
$2\pi/3, 0.216$	6	0.0397	0.0121 ± 0.0017	0.31 ± 0.04	0.056 ± 0.008	-0.031 ± 0.003
$\pi, 0.115$	8	0.0270	0.0062 ± 0.0004	0.229 ± 0.015	0.054 ± 0.004	-0.010 ± 0.002
$\pi, 0.220$	6	0.0534	0.0138 ± 0.0023	0.26 ± 0.04	0.063 ± 0.010	-0.015 ± 0.003
$2\pi, 0.127$	6	0.0709	0.0068 ± 0.0009	0.096 ± 0.013	0.053 ± 0.007	-0.005 ± 0.002
$2\pi, 0.173$	6	0.0802	0.0096 ± 0.0018	0.120 ± 0.023	0.055 ± 0.011	-0.004

Table 2: The six arms. Errors are block jackknives over runs; $\alpha' = 1 - \widehat{(1 - \alpha')}$ is the box-frame value, without the $T = 0$ pair-size baseline of the preceding paper (all values small and negative; the Iordanskii value would be $+\rho_n/\rho$).

FL2 fails. The coefficient is 0.31 ± 0.03 , 0.232 ± 0.014 , 0.102 ± 0.011 at $k_c\xi = 2.1, 3.1, 6.3$ (weighted over the arms at each cutoff): $(\max - \min)/\text{mean} = 0.98$, against the registered 0.2. **The rival fails too:** c/k_c is not constant ($\chi^2 = 204$ for 2 degrees of freedom) — the coefficient *falls* with the cutoff, the Born law makes it rise. **FL3's expectation fails** (spread 10–30 %). **One common coefficient fails** ($\chi^2 = 76$ for 5 degrees of freedom). **FL1 holds** at every cutoff where two temperatures exist ($2\pi/3$: 0.32, 0.31; π : 0.23, 0.26; 2π : 0.096 ± 0.013 , 0.12 ± 0.02) — each cutoff is proportional to its own ρ_n ; it is the coefficient that depends on the cutoff.

4 A temperature law, found after the fact

The comparison of the two panels of Fig. 2 is the finding. At fixed T the friction hardly moves with the cutoff, while $\rho_n/(\rho T)$ moves by a factor of three to four (0.15–0.18 at $2\pi/3$, 0.24 at π , 0.46–0.56 at 2π); $\alpha/T = 0.0540 \pm 0.0026$ in all six arms ($\chi^2 = 1.28$ for 5 degrees of freedom, against 76 for a common c). In the earlier paper the three temperatures were at one cutoff, where $\rho_n/\rho \propto T$ (we measured $\rho_n/(\rho T) = 0.23, 0.24, 0.27$), and a proportionality to either could not be told from the other; the cutoff scan breaks the degeneracy.

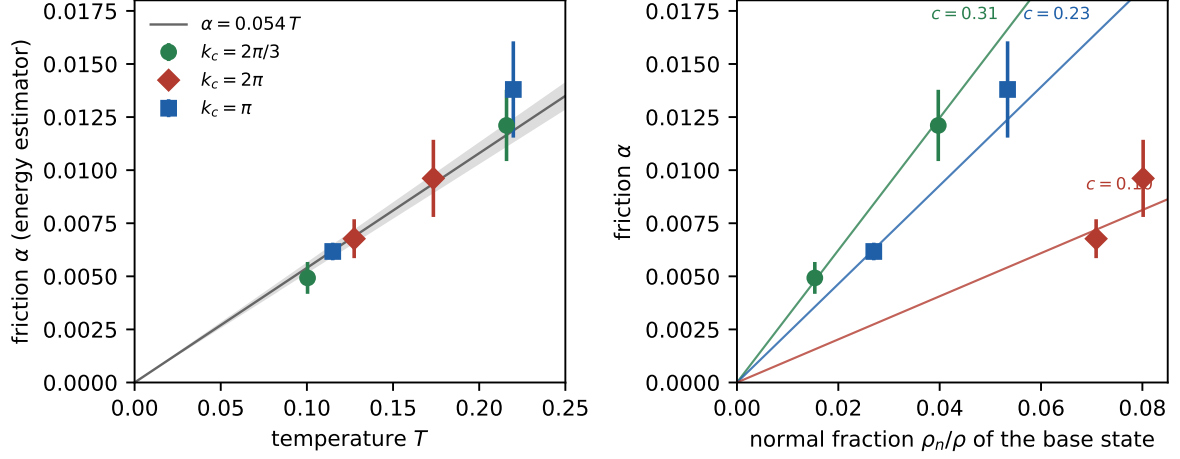


Figure 2: Left: the friction against temperature, for the three cutoffs; the line is $\alpha = 0.054 T$ (band: one standard error). Right: the same data against the normal fraction of the base state; the three straight lines through the origin have different slopes (Table 2). Open symbols: fewer than six runs.

Hypothesis H-T (post hoc; filed as amendment FL-A1 before the last arm was read): $\alpha = aT$, $a = 0.054 \pm 0.003$, independent of the cutoff for $k_c \xi \geq 2$ (a cutoff at 2.1 already carries the whole friction). **Prediction fixed in advance** for the arm running when it was written ($k_c = 2\pi$, $T = 0.1734$, $\rho_n/\rho = 0.0802$): H-T gives $\alpha_E = 0.0094 \pm 0.0010$; the proportionality to ρ_n at that cutoff's coefficient gives 0.0077; [0.0084, 0.0104] supports H-T, [0.0067, 0.0087] the alternative, the overlap is inconclusive. **Outcome** (read 2026-10-09 by the pre-written rule): $\alpha_E = 0.0096 \pm 0.0018$ ($\alpha_{\text{reg}} = 0.0110 \pm 0.0014$; six runs, two ended by annihilation, two by track loss), $c = 0.120 \pm 0.023$, $\alpha/T = 0.055 \pm 0.011$ — **in the window of H-T**, outside that of the ρ_n law (inclusion by the rule; by itself the arm lies 0.2σ from 0.0094 and 1.0σ from 0.0077).

Estimator robustness (found while porting the estimators to Rust). A re-implementation of the registered estimators in a second language, bit-identical to the Python on four real tracks, was run on the registered synthetic gate over many seeds and showed that the energy estimator of α , the primary estimator used throughout, is *biased low by the diffusion*: for a true $\alpha = 0.02$ it returns 0.0200 without diffusion and 0.0183, 0.0163, 0.0156 at $\eta = 5 \times 10^{-4}$, 10^{-3} , 2×10^{-3} (−8, −18, −22 %), independently of the detection noise, while the regression estimator is unbiased (0.0194); the registered gate passed on one seed. In the data the two estimators differ by 5–21 % in that direction. With the regression estimator $\alpha/T = 0.0598 \pm 0.0023$ ($\chi^2 = 5.2$ for 5 degrees of freedom; 0.0540 ± 0.0026 with the energy estimator) and the coefficient c is 0.355 ± 0.035 , 0.250 ± 0.012 , 0.124 ± 0.011 at the three cutoffs, a factor 2.9: every conclusion above stands, with absolute values of α higher by about 11 % (the values quoted for $k_c = \pi$ in the preceding paper are low by 5–20 %).

Why a temperature law is natural, and what it requires. In a classical field every mode carries the energy T , so occupations are $n_k = T/\varepsilon_k$ and anything linear in the occupation is linear in T . The kinetic form $D = \frac{1}{2} \int \frac{d^2 k}{(2\pi)^2} \frac{T}{\varepsilon_k^2} k^2 c_g \sigma_{\text{tr}}$ with the Bogoliubov energy $\varepsilon_k = \sqrt{k^2 + k^4/4}$ is then T times a function of the cutoff; for $k\xi \gg 1$ ($\varepsilon \simeq k^2/2$, $c_g \simeq k$) the integrand is $\propto \sigma_{\text{tr}}(k)$, so a drag that does not depend on the cutoff beyond $k_c \xi \simeq 2$ requires $\int^\infty \sigma_{\text{tr}} dk$ to converge — a transport cross-section falling faster than $1/k$ — whereas the same weights make ρ_n grow

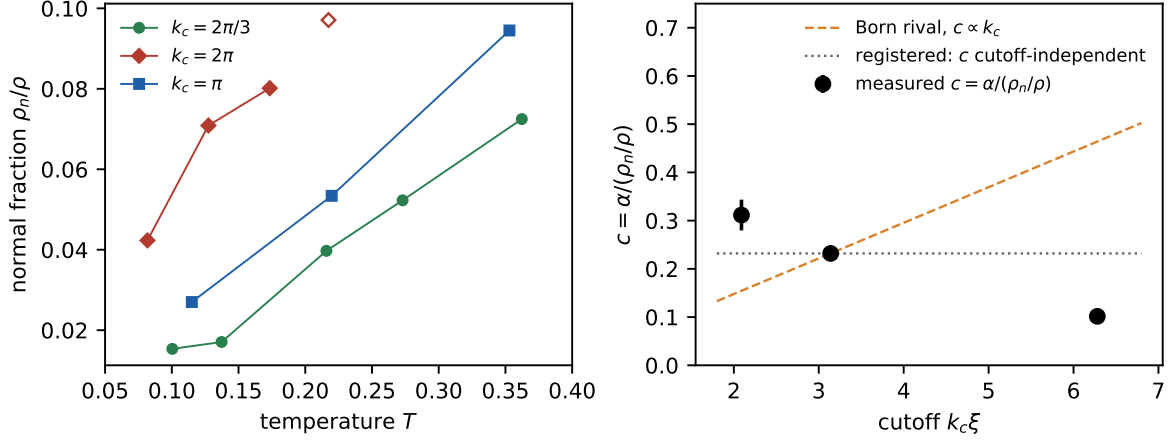


Figure 3: Left: the normal fraction against temperature at the three cutoffs (open: not admitted). Right: the coefficient c against the cutoff: the registered (constant) and rival ($\propto k_c$) predictions fail.

logarithmically with the cutoff. This is a constraint on the scatterer if the independent-mode picture holds; the long-wavelength Born cross-section $\sigma_{\parallel} = \kappa^2 k / 8c^2$ grows linearly with k and is valid for fewer than 3% of the modes of a cutoff π [3], so it cannot be extrapolated. The direct measurement of $\sigma_{\text{tr}}(k)$ — a monochromatic Bogoliubov wave scattered by one vortex at $T = 0$ — would settle it and is the obvious next experiment.

5 What this corrects

1. The statement “ $\alpha \simeq 0.24 \rho_n/\rho$ ” of the preceding paper [1] holds at $k_c = \pi$ only; the relation that survives the cutoff scan is $\alpha = 0.054 T$ (confirmed on the last arm by the registered rule).
2. The reading of 0.24 as a mode-averaged transport cross-section of the vortex, $\langle c_g \sigma_{\text{tr}} \rangle = 1.4 \xi$, “the geometric size of the core”, is withdrawn: the same quantity is 1.9, 1.4, 0.59 ξ at the three cutoffs. The identity of Appendix B stands; its coefficient is a property of the bath’s mode set, not of the vortex.
3. The recommendation to compare with quantum gases at equal ρ_n/ρ rather than equal T/T_c is withdrawn in the same sense: in this model ρ_n/ρ depends on the cutoff, α does not. (A comparison with experiment needs a map from the classical-field temperature to the physical one, and that map itself depends on the cutoff; the claim here is internal to the model.)

6 Limits

One box ($L = 64$), one interaction strength ($mg = 1$), pairs of two sizes and two or three temperatures per cutoff; the two new cutoffs at one and two temperatures; the dependence on the pair size (α larger for $d_0 = 12$ than for $d_0 = 8$ by 38–85% at the new cutoffs, not at π) and the small negative transverse coefficient, whose magnitude grows toward the coarse cutoff ($\alpha' = -0.005$ at 2π , -0.010 and -0.015 at π , -0.017 and -0.031 at $2\pi/3$; always far from the Iordanskii value) are reported and unexplained — either a bias of the energy estimator at the coarse cutoff or a finite-pair effect. The cutoff $2\pi/3$ shares its grid with π and the cutoff 2π has a finer grid: a lattice-spacing effect is not excluded by this design, though the $T = 0$ pair is insensitive to it to 0.3%. The energy estimator of α is biased low by diffusion (above), so the

absolute coefficient is 0.054–0.060; the law is estimator-independent in its form. The temperature law was formulated after five arms were read and confirmed by the rule on the sixth, whose error is large; it is a well-supported empirical relation over $T = 0.10$ – 0.22 and three cutoffs, not a derived one, and it has not been tested above $T = 0.22$ (where thermal vortices enter) or at other box sizes or interaction strengths.

7 Conclusion

A pre-registered test failed in the way that matters: the coefficient between the friction and the normal density is not a property of the vortex, and neither the registered cutoff-independence nor its Born rival describes it. What the test found instead is a friction that follows the temperature and ignores the cutoff over a factor three, which is a stricter statement than the one it replaces, and a correction of the preceding paper’s reading, made here rather than left to a reader.

A Registered items and outcomes

item	registered (before any run)	outcome
FL1	$\alpha \propto \rho_n$ at each cutoff	holds at all three cutoffs (each to its own ρ_n)
FL2	c within 20 % across cutoffs	fails (spread 0.98)
Born rival	$c \propto k_c$	rejected ($\chi^2 = 204/2$)
FL3	spread 10–30 %, $c(2\pi/3) \leq c(\pi) \leq c(2\pi)$	fails (ordering reversed)
$T = 0$ control	speed within 4 %, $ \alpha \leq 10^{-3}$	passes at all cutoffs; η floor marginal at $2\pi/3$
G0 energy criterion	α_E within 15 % (one numpy seed)	passes on 2 of 12 seeds: biased low by diffusion (regression unbiased)
base admission	raw $n_v < 0.5$	fine $T = 0.217$ base fails; replaced by $T = 0.173$ (admitted, $n_v = 0.08$)
H-T	post hoc, FL-A1	consistent ($\chi^2 = 1.28/5$); registered window for the last arm met (0.0096 in [0.0084, 0.0104])

B Lean

FrictionKinetic: for a finite set of modes with nonnegative weights w_k and drag factors $f_k = c_g \sigma_{tr}$, $D = \sum w_k f_k$ and $\rho_n = \sum w_k$ give $\alpha/(\rho_n/\rho) = \frac{\rho}{\rho_s \kappa} \cdot \sum w_k f_k / \sum w_k$ (an identity), the weighted mean lies between $\min f$ and $\max f$, and for equal weights (Rayleigh–Jeans with a phonon dispersion) it is the arithmetic mean. Nothing about the value of σ_{tr} or its dependence on k is asserted; that is the measurement.

C Code

The analysis (`exploration/pgpe/analyze_friction_law.py`) applies the registered estimators to every arm and writes the verdict file; the core of the cutoff comparison is

```
def wmean(v, s):
    v, s = np.array(v), np.array(s); w = 1 / s ** 2; m = (w * v).sum() / w.sum()
    return float(m), float(1 / np.sqrt(w.sum())), float(((v - m) ** 2 * w).sum()) # mean, se, chi2

c_by_cutoff = {k: wmean([x["c_energy"] for x in v], [x["c_energy_se"] for x in v]) for k, v in
    arms_by_cutoff.items()}
```

```
spread = (max(m for m, _, _ in c_by_cutoff.values()) - min(m for m, _, _ in c_by_cutoff.values())) /
    mean_of_means # FL2: < 0.20
m_T, se_T, chi2_T = wmean([a["alpha_energy_over_T"] for a in arms], [a["alpha_energy_over_T_se"] for a in
    arms]) # H-T
```

Data, scripts, pre-registrations and the ledger: [7].

References

- [1] Xavier Callens. Vortex transport and screening in a closed two-dimensional bose field (version 2.1). Zenodo, 2026. doi:10.5281/zenodo.23225293.
- [2] S. V. Iordanskii. On the mutual friction between the normal and superfluid components in a rotating bose gas. *Annals of Physics* 29, 335-349, 1964.
- [3] E. B. Sonin. Magnus force in superfluids and superconductors. *Phys. Rev. B* 55, 485, 1997. arXiv:cond-mat/9606099.
- [4] Geol Moon, Woo Jin Kwon, Hyunjik Lee, and Yong il Shin. Thermal friction on quantum vortices in a bose-einstein condensate. *Phys. Rev. A* 92, 051601(R), 2015. arXiv:1509.02236.
- [5] W. J. Kwon, G. Del Pace, K. Xhani, L. Galantucci, A. Muzi Falconi, M. Inguscio, F. Scazza, and G. Roati. Sound emission and annihilations in a programmable quantum vortex collider. *Nature* 600, 64-69, 2021. arXiv:2105.15180.
- [6] Vishwanath Shukla, Marc Brachet, and Rahul Pandit. Superfluid mutual-friction coefficients from vortex dynamics in the two-dimensional galerkin-truncated gross-pitaevskii equation. 2014. arXiv:1412.0706.
- [7] Xavier Callens. Socrateai-scientific-quantumfluids: Lean 4 library, pre-registrations and numerical record (v1.17.0). Zenodo, 2026. doi:10.5281/zenodo.23225129.